

Analyzing how snowpack variations affect the water quality at Lake Tahoe

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Background

Lake Tahoe is the largest alpine lake in North America

- ~ 200 sq. miles at 6000+ ft elevation
 - Crystal-clear
 - Recreation, fisheries, drinking water

Sierra Nevada snowpack is the dominant annual water source

- Each spring, snowmelt feeds into the lake
- Year-to-year snowpack magnitude and timing varies...
- Snowmelt may matter for water quality



Lake Tahoe and the Sierra Nevada snowpack

Our Study

How do inter-annual variations in the magnitude and timing of the Sierra Nevada snowpack melt affect the turbidity and chlorophyll content of Lake Tahoe from 2017-2025?



Dilution:

Freshwater melt reduces sediment and nutrient concentrations

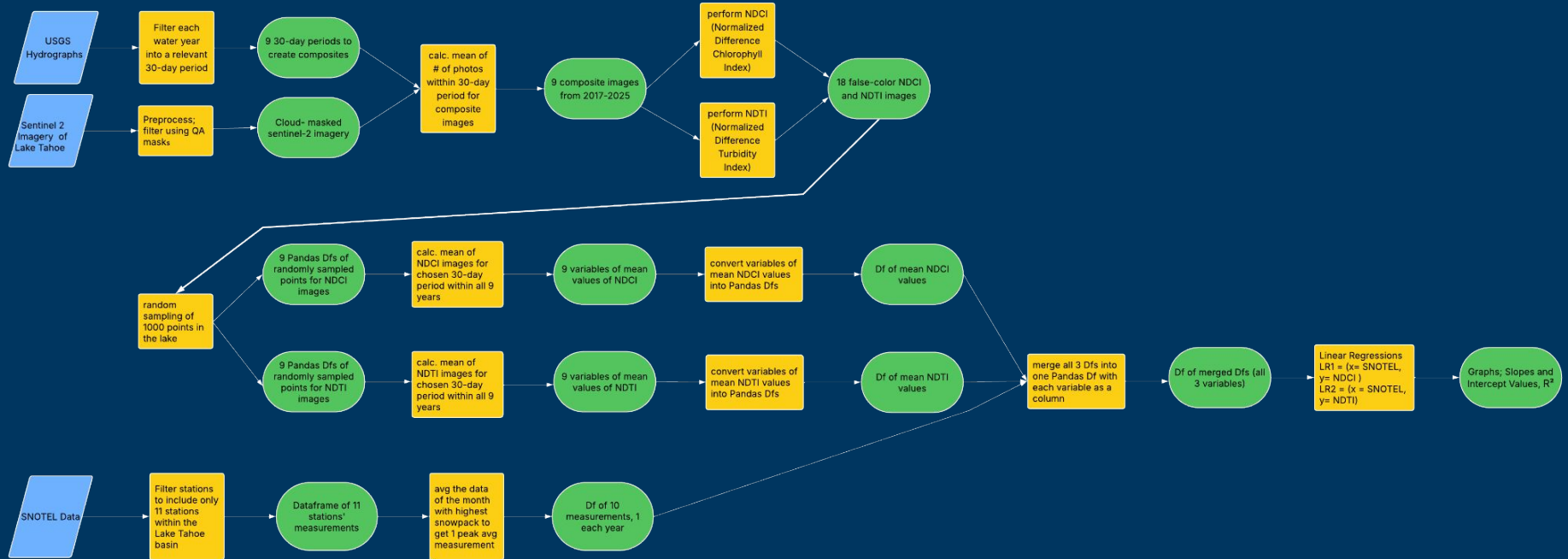
Loading:

Runoff carries in new sediments and nutrients into the lake

Net effect?

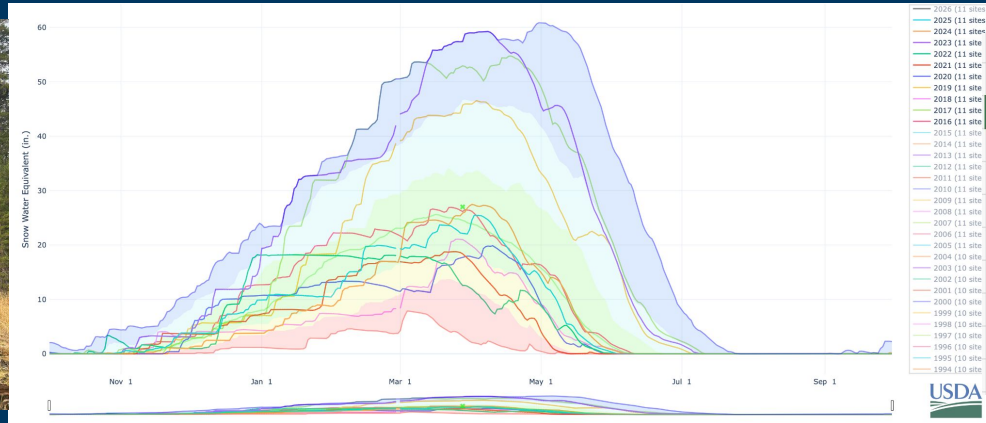
Hypothesized that more snow will lead to higher water quality

Workflow



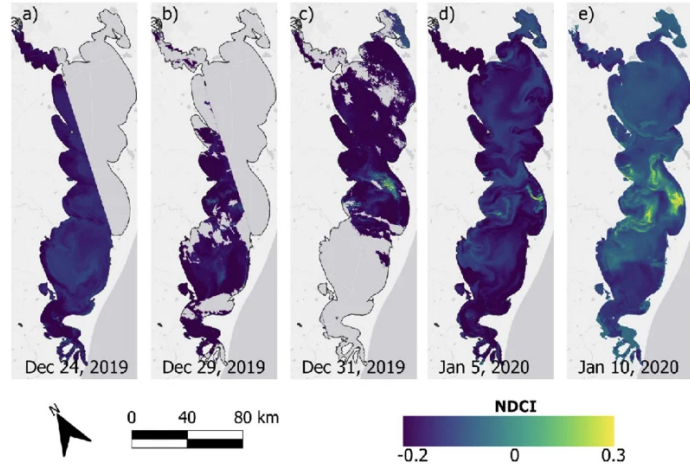
SNOTEL Stations

- Derived from USDA National Resource Conservation Service
- Filtered for Lake Tahoe, 11 SNOTEL sites, Basin charts combine data from multiple SNOTEL sites within and adjacent to a watershed.
- Found maximum snow water equivalent for each year (9 most recent years)
- 2016-2025

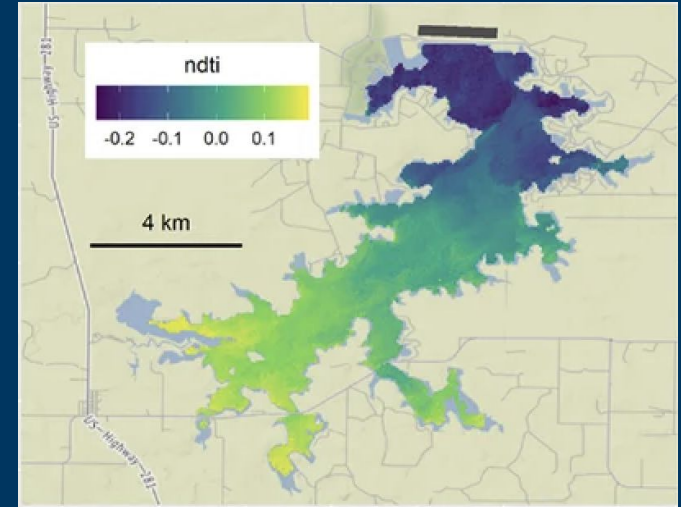


Season-Year	Peak Snow W Date	
2016	26.9	22-Mar
2017	54.7	17-Apr
2018	21.1	27-Mar
2019	46.5	3-Apr
2020	19.9	10-Apr
2021	18.8	25-Mar
2022	18.2	20-Dec
2023	59.2	4-Apr
2024	27.4	2-Apr
2025	25.5	4-Apr

NDCI and NDTI



$$NDCI = \frac{\text{Red Edge} - \text{Red}}{\text{Red Edge} + \text{Red}}$$



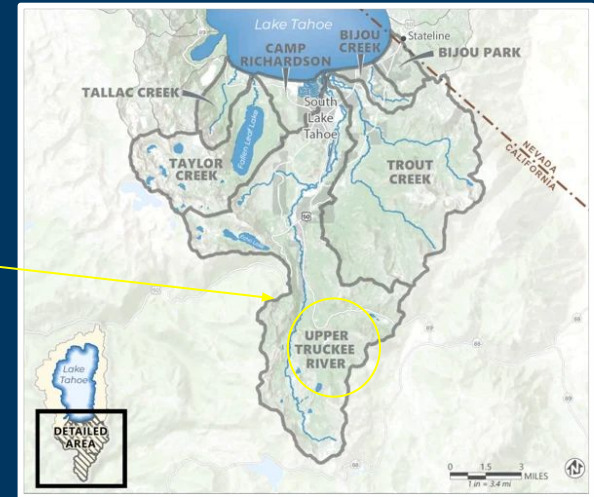
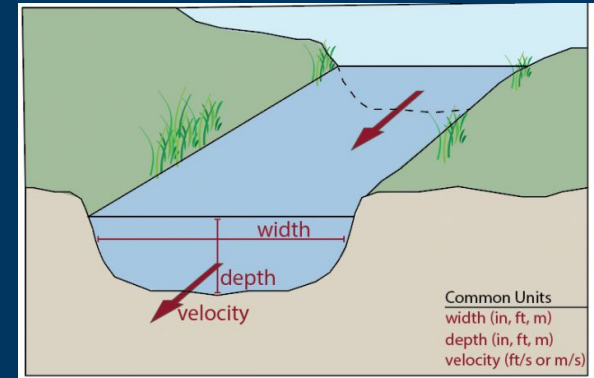
$$NDTI = \frac{\text{Red} - \text{Green}}{\text{Red} + \text{Green}}$$

Sentinel-2 Imagery

- Harmonized Sentinel-2 Multispectral Instrument, Level-2A (Surface Reflectance)
- Fine spectral resolution
 - Red edge band allows makes NDCI possible
- Only available from March 28, 2017 - Present
- 5 day revisit time
 - Up to 6 images per timeframe
- 10m resolution in RGB
 - NDTI
- 20m resolution for red edge and near infrared
 - NDCI

Methods - Snowmelt Periods

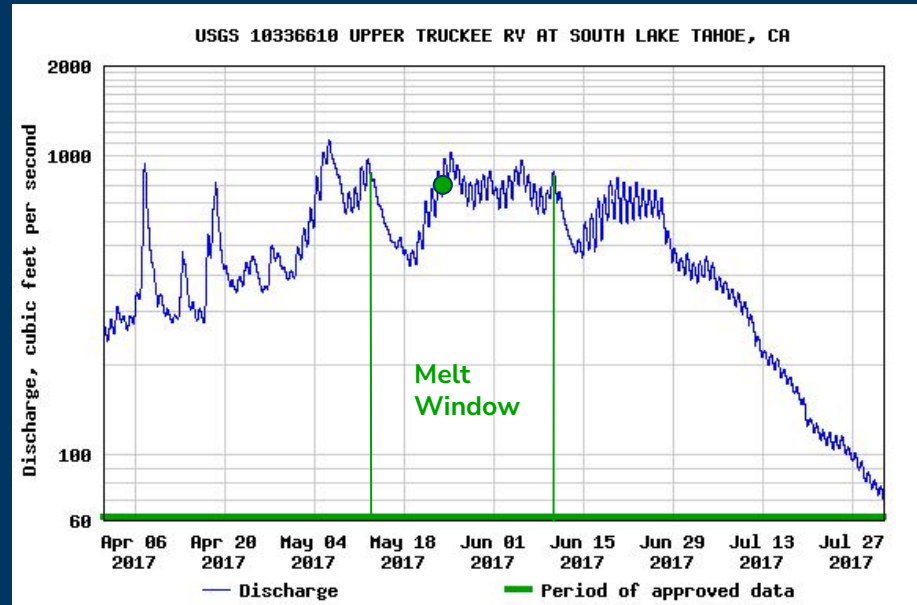
- Need: images to apply NDCI/NDTI to its composite ... a “Melt Window”
- Utilize discharge ($Q = A \times V$) ... (cfs)
- ~ 50% Lake Tahoe water comes through Upper Truckee River
- Need algorithm to find highest loading period...



Methods - Water Year 2017 Melt Window

Temporal Centroid of Streamflow (CT) Algorithm

- April 1-July 31
- 50% of total discharge = CT
- 30-day Window =
CT - 10 days,
CT + 20 days
- CT = May 22
- Window = May 12 - June 11



Methods - NDCI and NDTI

NDCI:

- Red Edge-Red / Red Edge+Red
 - B5-B4 / B5+B4 for Sentinel-2
- Blue to green gradient color palette
 - Blue shows low chlorophyll
 - Green shows high chlorophyll

NDTI:

- Red-Green / Red+Green
 - B4-B3 / B4+B3 for Sentinel-2
- White to purple gradient color palette
 - 'Purples' in GEE colormaps
 - White shows low turbidity
 - Purple shows high turbidity

```
# NDCI Expression
ndci_expression = '(redEdge - Red) / (redEdge + Red)'

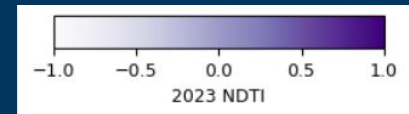
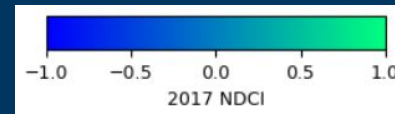
# NDTI Expression
ndti_expression = '(Red - Green) / (Red + Green)'

palette_NDCI = cm.get_palette('winter', 20)
palette_NDTI = cm.get_palette('Purples', 20)

ndci_vis = {
    'min': -1,
    'max': 1,
    'palette': palette_NDCI
}

ndti_vis = {
    'min': -1,
    'max': 1,
    'palette': palette_NDTI
}
```

Min and Max are -1 and 1 because they are indices. 20 bins for each palette allows for 1 shade for each 0.1 interval



Methods - Composite Images

- Lake Tahoe GEOJSON converted to Earth Engine geometry
- Cloud masked using QA band
 - Earth Engine sample code
- Created a function, `make_index_maps`, that creates NDCI and NDTI mean composites
 - Returns an interactive map with a layer for each index
- Makes 9 interactive maps using 'for' 'in' operations

```
def make_index_maps(start_date, end_date, output_name):

    dataset = (
        ee.ImageCollection('COPERNICUS/S2_SR_HARMONIZED')
        .filterDate(start_date, end_date)
        # Pre-filter to get less cloudy granules.
        .filter(ee.Filter.lt('CLOUDY_PIXEL_PERCENTAGE', 20))
        .map(mask_s2_clouds)
    )

    #Clip dataset to Study Site
    MeanComposite = dataset.mean().clip(LakeTahoe_Geom)
    ndci_image = MeanComposite.expression(ndci_expression, {'redEdge': MeanComposite.select('B5'), 'Red': MeanComposite.select('B4')})
    ndti_image = MeanComposite.expression(ndti_expression, {'Red': MeanComposite.select('B4'), 'Green': MeanComposite.select('B3')})

    #Create Map

    m = geemap.Map()
    m.set_center(-120.044861, 39.089568, 11)
    m.add_layer(ndci_image, ndci_vis, output_name + ' NDCI')
    m.add_layer(ndti_image, ndti_vis, output_name + ' NDTI')
    m.add_colorbar(ndci_vis, label= output_name + ' NDCI', orientation='horizontal', position='bottomleft')
    m.add_colorbar(ndti_vis, label= output_name + ' NDTI', orientation='horizontal', position='bottomright')
    m
    return m
```

```
interactive_map = {}
for start_date, end_date, output_name in timeframes:
    interactive_map[output_name] = make_index_maps(start_date, end_date, output_name)
```

```
for year, map_obj in interactive_map.items():
    print(f"Map for Year: {year}")
    display(map_obj)
```

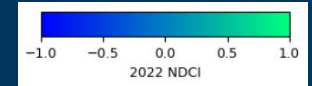
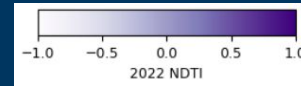
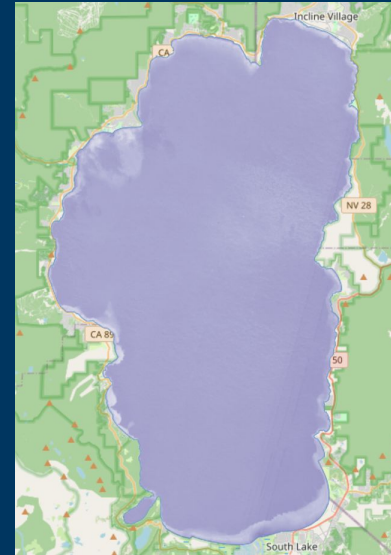
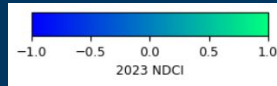
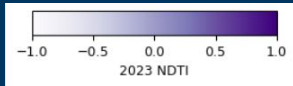
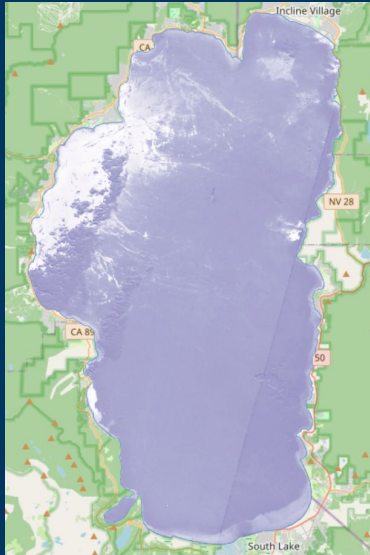
Methods - Analysis

- Sampled 1000 pixels from each image and created dataframes
 - Created a function called `get_sample_points`
- Found the mean NDTI and NDCI from the samples
 - Merged with snowpack data
- Scatter plots with linear regression lines
- Snowpack size on X axis
- NDCI or NDTI on Y axis
- Each point is labeled with its respective year (9 points per graph)

```
ndci_sample_points = ndci_image.sample(  
    region = Lake_Tahoe_Geom,  
    scale = 20,  
    numPixels = 1000,  
    seed = 0,  
    geometries = False  
)  
ndti_sample_points = ndti_image.sample(  
    region = Lake_Tahoe_Geom,  
    scale = 10,  
    numPixels = 1000,  
    seed = 0,  
    geometries = False
```

An excerpt of code from the function `get_sample_points`. The seed is set to 0 so that the random sample is reproducible.

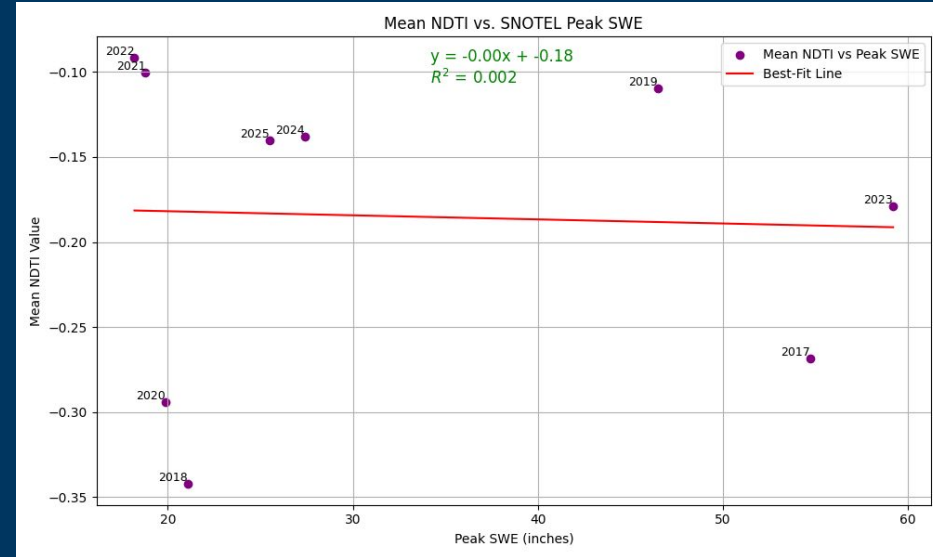
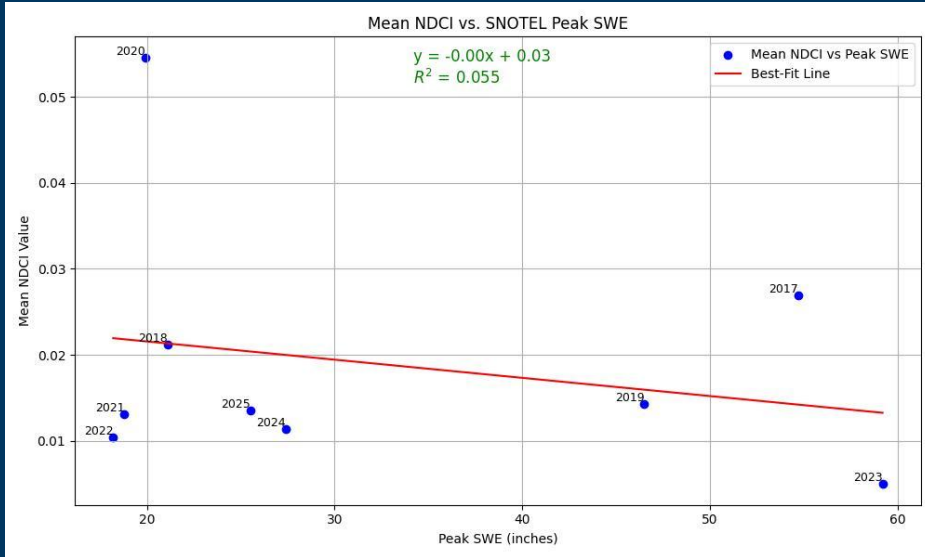
Results



2023 NDCI and NDTI Images
Largest Snowpack

2022 NDCI and NDTI Images
Smallest Snowpack

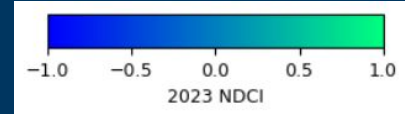
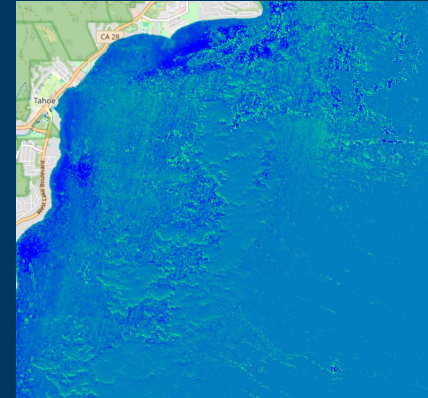
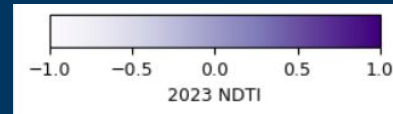
Results



- There is no relationship between peak snowpack levels and NDCI or NDTI
- Our result does not support our hypothesis

Limitations - NDTI and NDCI

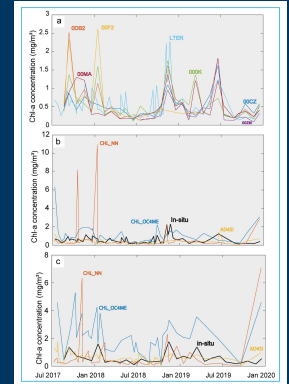
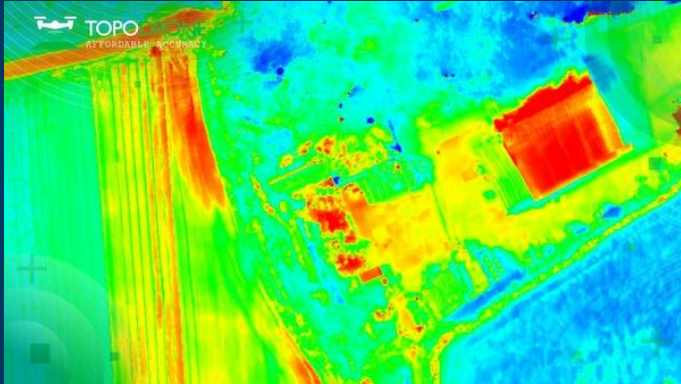
- NDTI is great for turbid waters such as rivers
 - High sediment load
- Does not account for algal growth
 - Especially in clear water with low sediment load
- Water that is less pure because of algae will show up as clear with NDTI
- Shallow water and structures (docks) show up as a higher turbidity level, and lower chlorophyll content



Areas with a high green reflectance, but low red reflectance means there is low sediment and turbidity, but in clear alpine water, it also means there is photosynthetic growth.

Future Directions

- Using different variables for outcomes
 - Independent variables (e.g. thermal imagery)
 - Dependent variables (e.g. sediment concentrations, in-situ data)
- Using different dates and time-periods for our variables' analyses
- Compute multivariate linear regressions



Conclusion

- Measured snowpack size in Lake Tahoe Basin using SNOTEL
- USGS Hydrographs to determine snowmelt windows
- Sentinel-2 imagery to analyze chlorophyll and turbidity in Lake Tahoe
 - How snowpack size affected these water quality metrics
- Found no strong relationship between snowpack size in the Lake Tahoe Basin and chlorophyll concentration and turbidity metrics

End